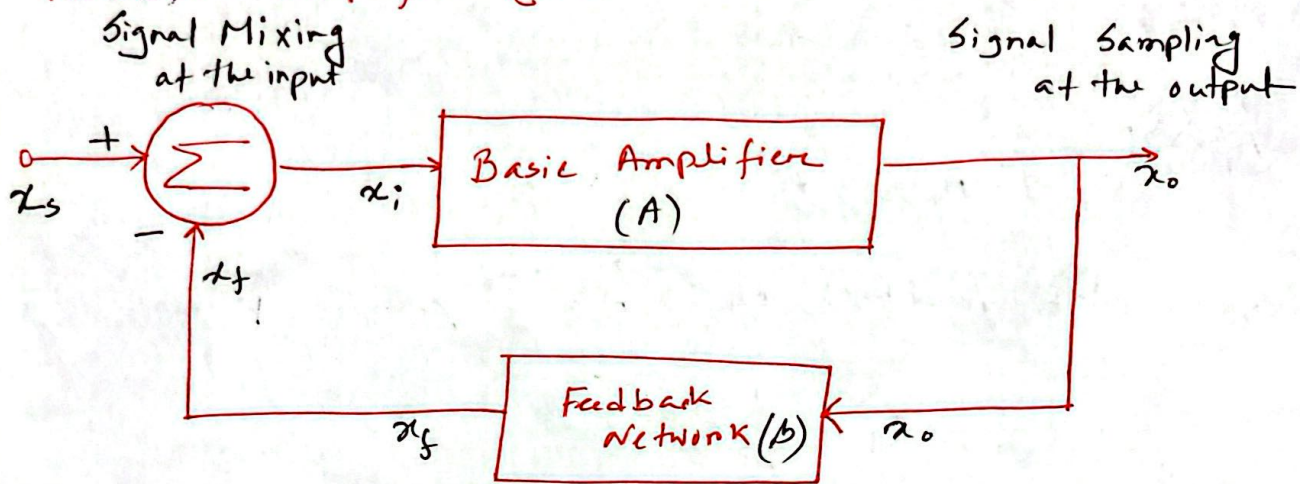


Feedback:

- In his search for methods for the design of an amplifier with stable gain for use in telephone repeaters, Harold Black, an electronics engineer with the Western Electric Company, invented the feedback amplifier in 1928.
- All electronic circuits for practical application will have some form of feedback, either implicit or explicit.
- It is also used in biological science.

Feedback Amplifier (Basic Structure):



Signal flow diagram of negative feedback (FB) Amp.

$$x_i = x_s - x_f \rightarrow \text{Difference Signal}$$

$$x_o = A x_i \rightarrow \text{Output Signal}$$

$$A = \frac{x_o}{x_i} \rightarrow \text{Gain of Basic Amp}$$

$$x_f = \beta x_o \rightarrow \text{Feedback Signal}$$

$$\beta = \frac{x_f}{x_o} \rightarrow \text{Feedback Factor}$$

$$A_f = \frac{x_o}{x_s} \rightarrow \text{Gain with feedback}$$

$$x_i = x_s - x_f$$

$$= x_s - \beta x_o \quad x_f = \beta x_o$$

$$= x_s - \beta A x_i \quad x_o = A x_i$$

$$x_s = x_i + \beta A x_i = x_i (1 + \beta A)$$

Gain with feedback,

$$A_f = \frac{x_o}{x_s} = \frac{A x_i}{x_i (1 + \beta A)}$$

$$\boxed{A_f = \frac{A}{1 + A\beta}}, \quad A_f < A.$$

Gain with feedback

If, $A\beta \gg 1$,

$$1 + A\beta \cong A\beta$$

$$\therefore \text{Gain with feedback, } A_f = \frac{A}{A\beta} = \frac{1}{\beta}$$

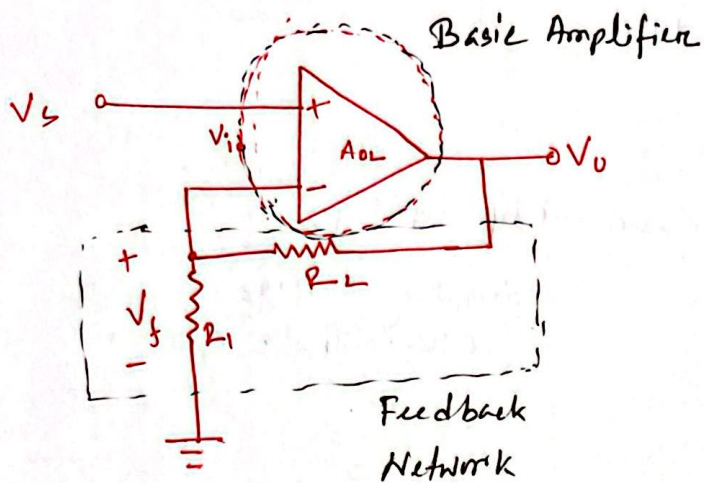
$$\therefore \boxed{A_f = \frac{1}{\beta}}$$

Gain of the FB Amplifier is independent of the gain (A) of the basic amplifier.

Benefits of negative FB Amp:

- Stabilizes gain
- widens bandwidth
- reduces noise
- reduces non-linear distortion.
- adjust input/output resistance.

Example of negative feedback amp →



$$V_{id} = V_s - V_f \quad \text{Difference Signal}$$

$$A_{OL} = \frac{V_o}{V_{id}}, \quad \text{Gain of Basic Amplifier}$$

$$\beta = \frac{V_f}{V_o}, \quad \text{Feedback Factor.}$$

Gain with feedback,

$$A_f = \frac{V_o}{V_s} = \frac{1}{\beta},$$

$$\text{For, } A_{OL} \beta \gg 1$$

$$\beta = \frac{V_f}{V_o} = \frac{R_1}{R_1 + R_L} \quad \text{feedback factor.}$$

$$A_f = \frac{1}{\beta} = \frac{R_1 + R_L}{R_1} = 1 + \frac{R_L}{R_1}$$

= A_{CL} , Close loop gain of non-inverting Amp.

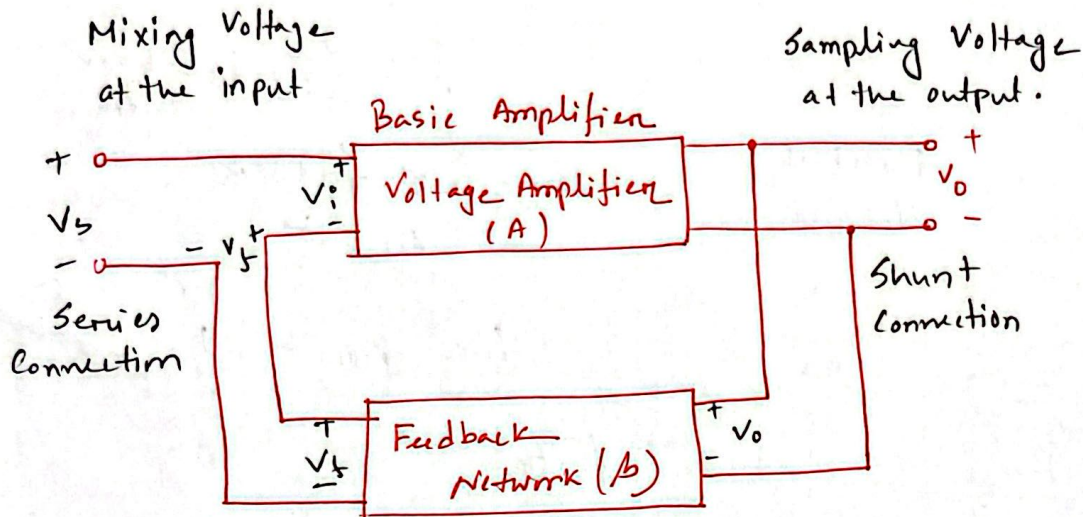
Independent of the gain (A_{OL}) of the basic amplifier.

Feedback Topologies:

- ① Voltage sampling → Sampling voltage at the output, Voltage mixing (Series-shunt) also mixing voltage at the input.
- ② Voltage Sampling → Sampling voltage at the output, Current Mixing (shunt-shunt) mixing current at the input.
- ③ Current Sampling → Sampling current at the output, Current Mixing (shunt-series) mixing voltage current at the input.

(4) Current Sampling $\rightarrow$ Sampling current at the output,
Voltage Mixing (series-series) mixing voltage at the input.

Voltage Sampling Voltage Mixing FB topology:
(Series-Shunt FB topology)

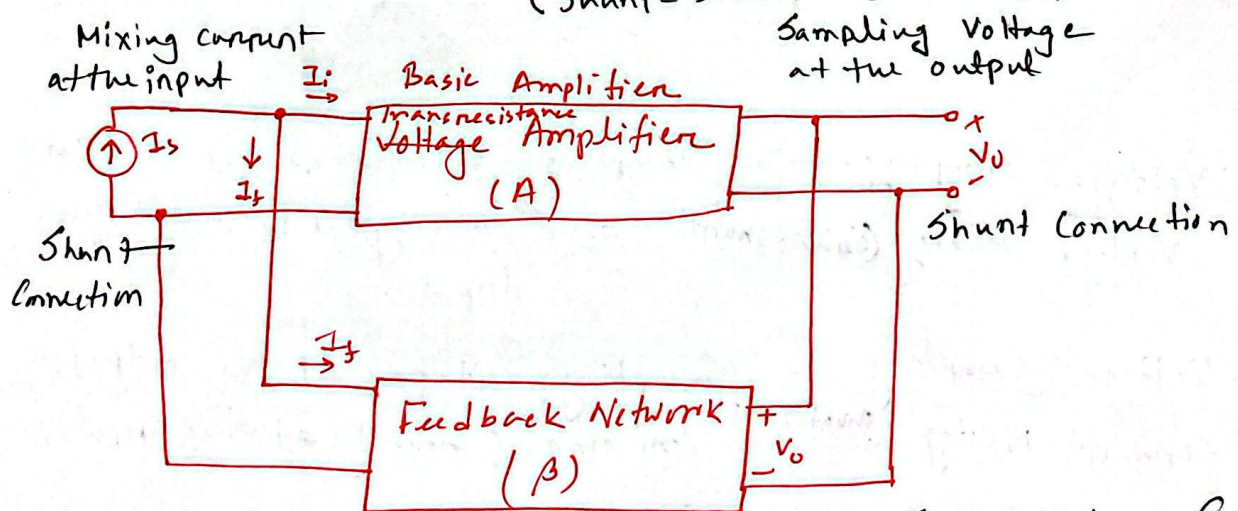


Gain of the basic Amp.

$$A = \frac{V_o}{V_i}, \text{ voltage gain}$$

Amplifier type $\rightarrow$ Voltage Amplifier.

Voltage Sampling Current Mixing FB topology:
(Shunt-Shunt FB topology)



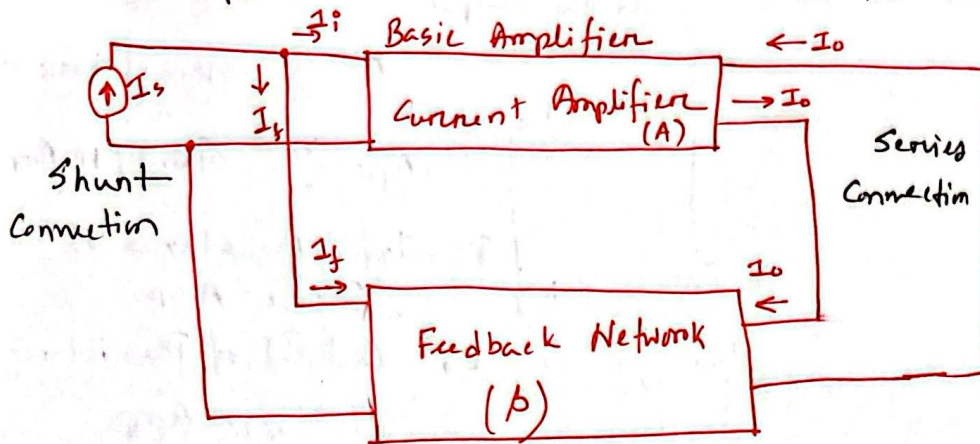
Gain of the Basic Amp - $A = \frac{V_o}{I_i}$, Transresistance Gain

Amplifier type: Transresistance Amp.

Current Sampling Current Mixing FB topology (Shunt-Series FB topology) ⁽³⁾

Mixing Current at the input

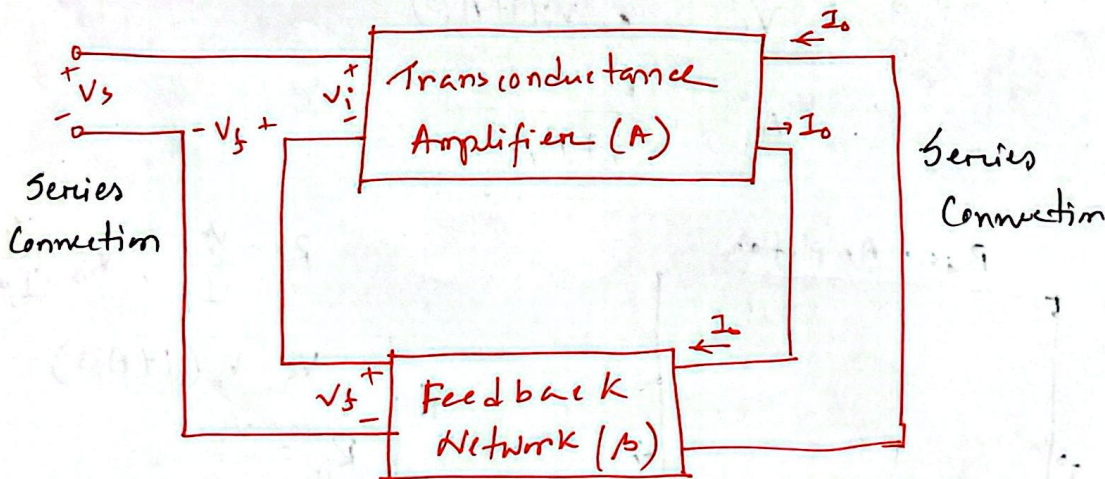
Sampling Current at the output



Gain of the Basic Amp, $A = \frac{I_o}{I_i}$, Current Gain

Amplifier type $\rightarrow$ Current Amplifier.

Current Sampling Voltage Mixing FB topology: (Series-Series FB topology)



Gain of the basic Amp.

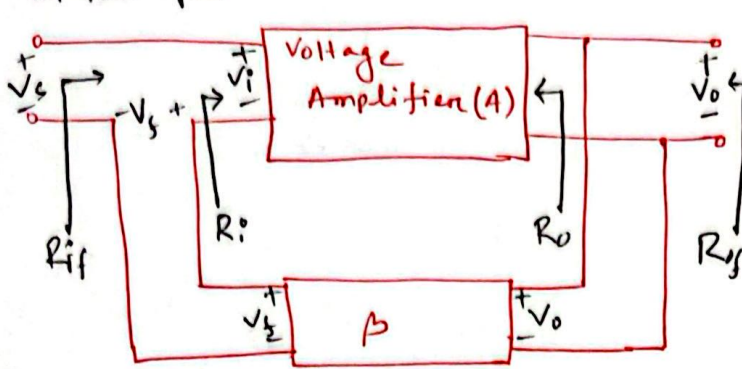
$A = \frac{I_o}{V_i}$, Transconductance gain

Amplifier type - Trans conductance Amp.

① Voltage Sampling Voltage Mixing FB topology: (series-shunt)

Mixing Voltage
at the input

Sampling Voltage
at the output.



$$A = \frac{V_o}{V_i} \text{ , Gain of basic amp}$$

$$A_f = \frac{V_o}{V_s}, \text{ Gain of FB Amp}$$

R_i = Input Resistance of Basic Amp

R_o = Output Resistance
of Basic Amp

R_{is} - Input Resistance of FB Amp

$R_{of} \rightarrow$ Output Resistance of FB Amp.

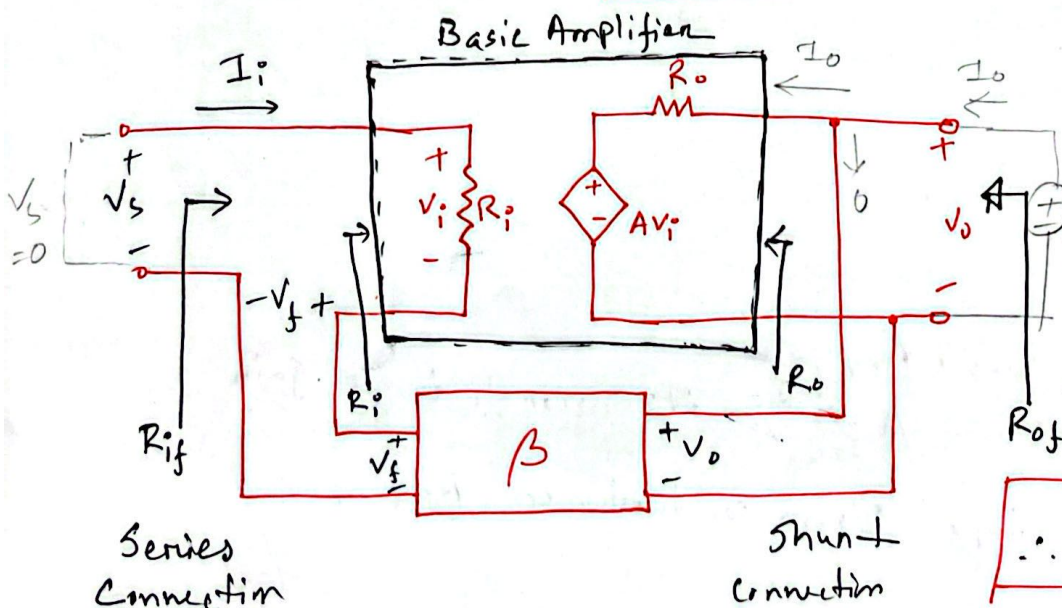
$$V_i = V_s - V_f = V_s - \beta V_o \quad [\because \beta = V_f/V_o]$$

$$= V_s - \beta A V_i \quad [A = v_o/v_i]$$

$$v_s = v_i + \beta A v_i = v_i (1 + A\beta)$$

$$\therefore A_f = \frac{V_o}{V_s} = \frac{AV_i}{V_i(1+A\beta)}$$

$$A_f = \frac{A}{1 + A\beta}$$



$$R_i = \frac{V_i}{I_i}, \quad R_{is} = \frac{V_s}{I_s}$$

$$V_s = V_i (1 + A\beta)$$

$$= \frac{V_s}{I_i} = R_i (1 + A\beta)$$

$$\therefore R_f = R_i(1 + A\beta)$$

$$R_{of} = \frac{V_t}{I_o}$$

$$V_t = V_o = I_o R_o + A V_i$$

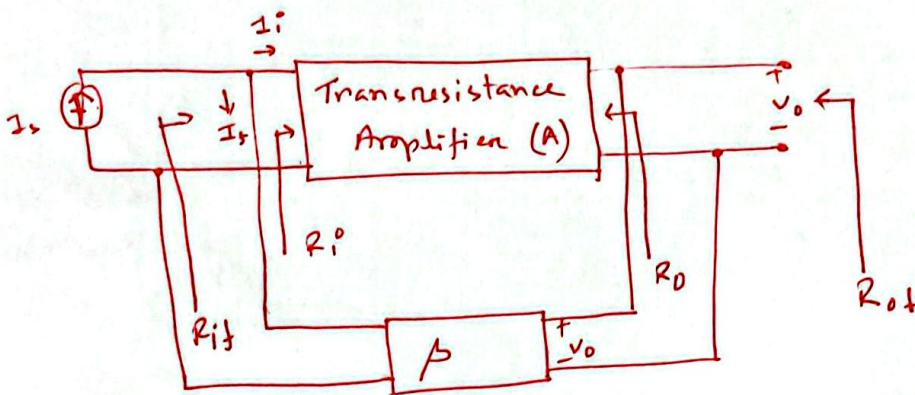
$$V_i = -V_f = -\beta V_o = -\beta V_t$$

$$V_t = I_o R_o - A \beta V_t$$

$$I_o R_o = V_t (1 + A \beta)$$

$$\therefore R_{of} = \frac{V_t}{I_o} = \frac{R_o}{1 + A \beta}$$

Voltage Sampling Current Mixing FB topology: (shunt-shunt)



$$A_f = A = \frac{V_o}{I_i}$$

$$\beta = \frac{I_f}{V_o}$$

$$I_i = I_s - I_f = I_s - \beta V_o$$

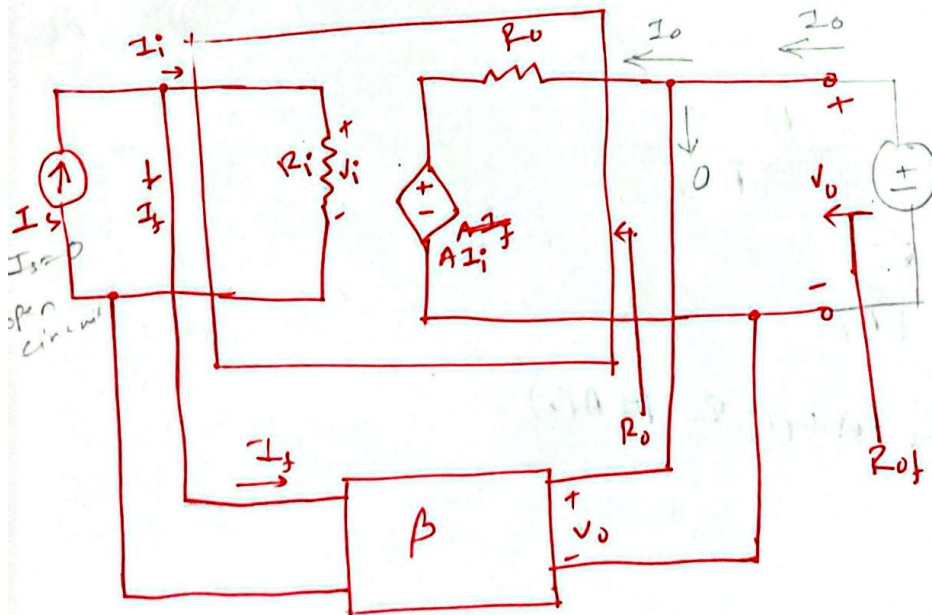
$$= I_s - \beta A I_i$$

$$I_s = I_i + \beta A I_i$$

$$= I_i (1 + A \beta)$$

$$\therefore A_f = \frac{V_o}{I_s} = \frac{A I_i}{I_i (1 + A \beta)}$$

$$\therefore A_f = \frac{A}{1 + A \beta}$$



$$R_i = \frac{V_i}{I_i}, R_{if} = \frac{V_o}{I_s}$$

$$I_s = I_i (1 + A \beta)$$

$$R_{if} = \frac{V_o}{I_s}$$

$$= \frac{V_o}{I_i (1 + A \beta)}$$

$$= \frac{R_i}{1 + A \beta}$$

$$\therefore R_{if} = \frac{R_i}{1 + A \beta}$$

$$R_{of} = \frac{V_t}{I_o}$$

$$V_t = V_o = I_o R_o + A I_i$$

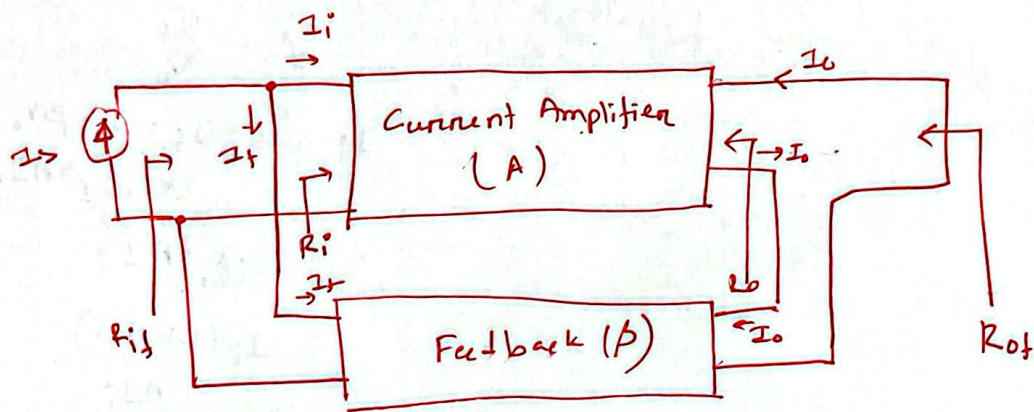
$$I_i = -I_f = -\beta V_o = -\beta V_t$$

$$V_t = I_o R_o - A \beta V_t$$

$$I_o R_o = V_t (1 + A \beta)$$

$$R_{of} = \frac{V_t}{I_o} = \frac{R_o}{1 + A \beta}$$

Current Sampling Current Mixing: (Shunt-Series)



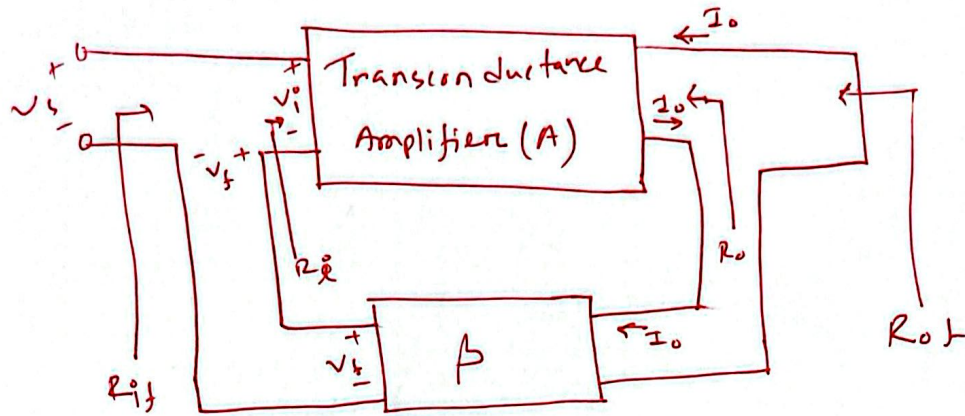
$$A = \frac{I_o}{I_i}$$

$$A_f = \frac{I_o}{I_s} = \frac{A}{1 + A \beta}$$

$$R_{if} = \frac{R_i}{1 + A \beta}$$

$$R_{of} = R_o (1 + A \beta)$$

Current Sampling Voltage Mixing: (Series-Shunt)



$$A = \frac{I_o}{V_i}$$

$$A_f = \frac{I_o}{V_s}$$

$$R_{if} = R_i (1 + A\beta)$$

$$R_{of} = R_o (1 + A\beta)$$